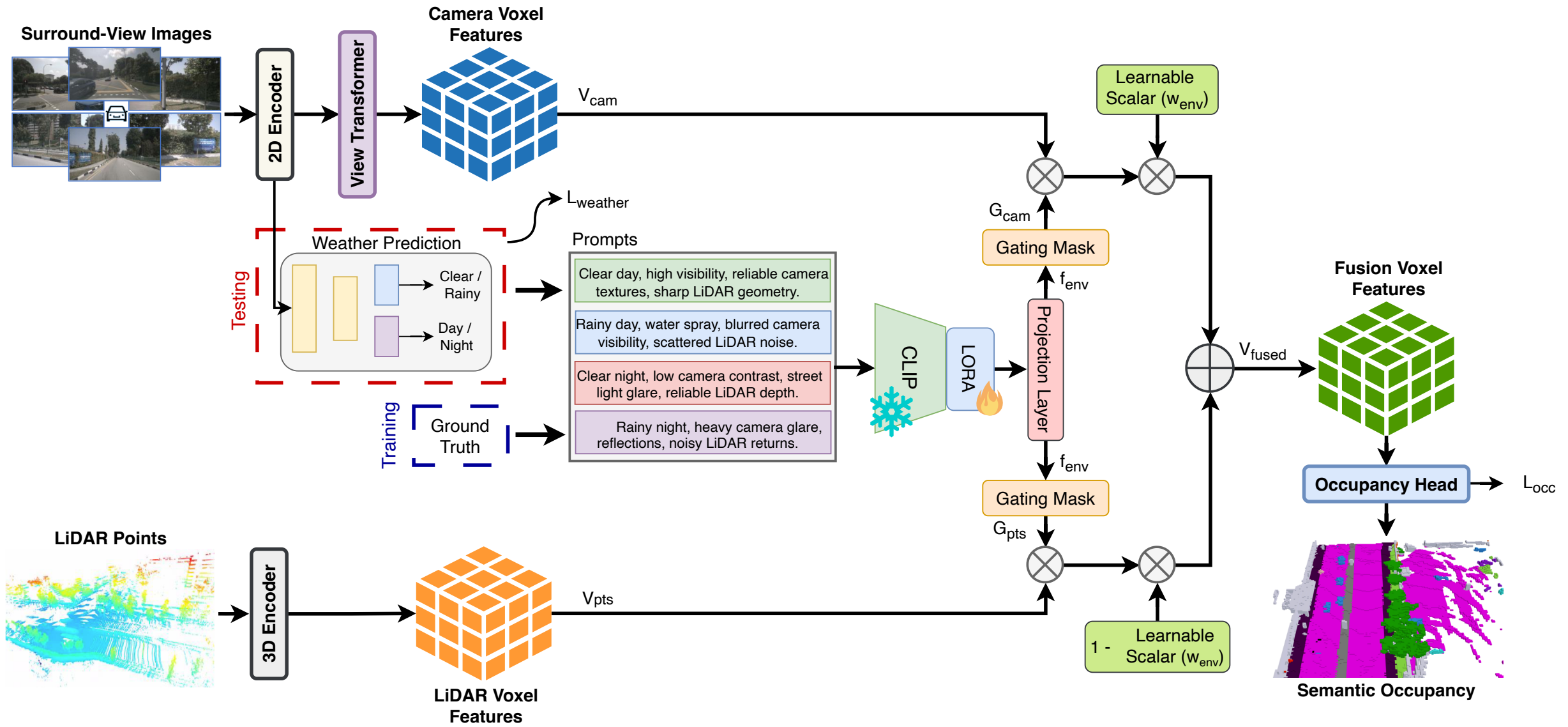
Surround-View Images
2D Encoder
View Transformer
Camera Voxel Features
V_{cam}
Learnable Scalar (w_{env})
G_{cam}
Gating Mask
f_{env}
Projection Layer
f_{env}
Gating Mask
G_{pts}
1 - Learnable Scalar (w_{env})
$L_{weather}$
Weather Prediction
Clear / Rainy
Day / Night
Testing
Training
Ground Truth
Prompts
Clear day, high visibility, reliable camera textures, sharp LiDAR geometry.
Rainy day, water spray, blurred camera visibility, scattered LiDAR noise.
Clear night, low camera contrast, street light glare, reliable LiDAR depth.
Rainy night, heavy camera glare, reflections, noisy LiDAR returns.
CLIP
LORA
V_{fused}
Fusion Voxel Features
Occupancy Head
L_{occ}
Semantic Occupancy
LiDAR Points
3D Encoder
V_{pts}
LiDAR Voxel Features